The set-theoretic rank of an ordinal equals the ordinal itself

The Open Logic Project contributors

openlogic:set-theory/spine:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-spine/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the induction framing and the three-term equality chain are verbatim; the author's superseded proof block survives only as source comments.

Editorial note. The label ordsetrankalpha names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own cor environment in the unit source.

Allowed standard prerequisites (editorial).

- Rank as strict supremum of ranks of members. Same unit, \ollabel{ranksupstrict} (source line 639). Final line of the proof (660): evaluates rank(alpha) as the strict supremum of the ranks of its members.
- Successor and limit case division, ordinal induction. Ordinals unit background: transfinite induction on ordinals. The proof opens 'Suppose for transfinite induction' at line 655.

Source reference. Original reference rankmemberslower: sources/olp-set-theory-spine/pinned-original-source.tex, source lines 602–602. Prerequisite pointer invoked at proof line(s) 662; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ranksupstrict: sources/olp-set-theory-spine/pinned-original-source.tex, source lines 639–639. Prerequisite pointer invoked at proof line(s) 659; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ordissetofsmallerord: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 447–447. Named cross-unit citation: the target label is staged in the corpus under unit olp-set-theory-ordinals

at source line 447; the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference Valphabasicprops: sources/olp-set-theory-spine/pinned-original-source.tex, source lines 847–847. Prerequisite pointer invoked at proof line(s) 670; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Corollary 0.1. *For any ordinal* α, $\operatorname{rank}\alpha = \alpha$.

Proof. Suppose for transfinite induction that $\operatorname{rank}\beta = \beta$ for all $\beta \in \alpha$. Now $\operatorname{rank}\alpha = \subsetneq_{\beta\in\alpha} \operatorname{rank}\beta = \subsetneq_{\beta\in\alpha} \beta = \alpha$ by ranksupstrict. □